

Aryan Shete

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Systems-focused engineer with hands-on experience in C/C++, Linux server deployment, low-level networking, and open-source contribution. Comfortable working close to the hardware and the OS.

Education

Bachelor of Technology — Electronics and Computer Science
Shah & Anchor Kutchhi Engineering College (SAKEC), Mumbai University

Expected May 2027

Technical Skills

Languages: C, C++, Python, Bash

Systems & OS: Linux (Arch, Ubuntu), POSIX syscalls, iptables, nftables, socket programming, process management, file descriptors, signals

Tools: Git, Neovim, GDB, AWS EC2, Docker, Codeforces (active)

Embedded: AVR microcontrollers, GPIO, interrupts, timer registers, bit manipulation

Projects

BombSquad Modded Server

C++ · Python · AWS EC2 · Linux · iptables

github.com/HeyFang/bombsquad-modded-server-scripts | Contributor to upstream Ballistica engine

- Forked and modified the Ballistica C++ game engine at both the C++ layer and Python scripting layer; contributed patches accepted into the upstream open-source repo.
- Deployed and maintained on AWS EC2 for several years; implemented network-level DOS mitigation using iptables rate-limiting and connection throttling rules to handle spoofers and flood attacks.
- Built IP-based and token-based admin authentication, structured player logging (IP, device UUID, session tokens) for moderation tooling, and custom chat commands; server files adopted by multiple community hosters.
- Active community of ~900 members on Discord; codebase kept clean, modular, and continuously updated.

Genesis

Bash · Docker · Ubuntu · Linux hardening

github.com/HeyFang/genesis-tool

- Menu-driven Bash automation tool for bootstrapping fresh Ubuntu servers: security hardening (SSH config, firewall rules, fail2ban), system monitoring setup, and Docker deployment — reducing manual setup from hours to minutes.

net-reason

Python · Linux · Bash · JSON

github.com/HeyFang/net-reason

- CLI diagnostic tool (netreason diagnosis) for triaging network and system errors on Linux/Ubuntu; uses a JSON-backed decision tree to narrow down root cause via interactive Y/N prompts and surface actionable fixes.

Security Surveillance Turret

C · AVR · Embedded · Hardware

github.com/HeyFang/security-turret

- Interrupt-driven embedded surveillance system on AVR: PIR sensor wakes the MCU from low-power idle, servo pans an ultrasonic sensor across 180°, locks onto targets within range using bit-manipulated timer registers — no delay() calls, full state-machine design.

Open Source

- **Ballistica Engine (Github)** — accepted contributor to the upstream C++/Python game engine repository; patches reviewed and merged by core maintainers. And contributed in few other ballistica related repositories.

Competitive Programming

- Active on Codeforces — solving div. 3 problems, progressing toward div. 2; focus on algorithmic problem solving in C++.